

Curriculum Vitae

Donggyu Lee

Korea Advanced Institute for Science and Technology

RESEARCH INTEREST

ML for Economics & Finance; Computational Social Science; Satellite Imagery & Remote Sensing

EDUCATION

Korea Advanced Institute for Science and Technology, Daejeon, South Korea

Ph.D. in Data Science

Sep 2026 –

- Advisor: Prof. Jihee Kim

Korea Advanced Institute for Science and Technology, Daejeon, South Korea

M.S. in Data Science

Sep 2024 – Aug 2026

- Advisor: Prof. Jihee Kim

Korea Advanced Institute for Science and Technology, Daejeon, South Korea

B.S. in Computer Science (Minor in Economics)

Mar 2017 – Aug 2024

- **Exchange Student**, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
Aug 2022 – Feb 2023

PUBLICATIONS

From Simulated Citizens to Simulated Deliberation: Challenges in Representation and Interaction

NeurIPS 2026 SocialAgent Workshop

2026

- **Authors:** Chaemin Jang, Junsik Min, Jaewoo Choi, **Donggyu Lee**, Haiin Lee, Junyoung Park, Namhee Kim, Hyunwoo Kim, Jungwon Kim, Juho Kim, Nuri Kim, Jihee Kim

Ideological Bias in LLMs' Economic Causal Reasoning

COLM 2026

2026

- **Authors:** **Donggyu Lee**, Hyeok Yun, Jungwon Kim, Junsik Min, Sungwon Park, Sangyoon Park, Jihee Kim

Tracking Economic Disparities in North Korea: Satellite Imagery Analysis Using AI with Policy Insights

ACM GOODIT 2025

2025

- **Authors:** Sumin Lee, Hyoshin Kim, Donghyun Ahn, **Donggyu Lee**, Jungwon Kim, Hyeok Yun, Chaemin Jang, Dongman Lee, Meeyoung Cha, Jihee Kim

LLM-Driven Socio-Economic Estimation for Visegrád Countries

ACM GOODIT 2025

2025

- **Authors:** Donghyun Ahn, Sumin Lee, **Donggyu Lee**, Jungwon Kim, Sungwon Han, Seongeon Lee, Younhyung Chae, Jihee Kim, Meeyoung Cha

WORKING PAPERS **EconCausal: A Context-Aware Causal Reasoning Benchmark for Large Language Models in Social Science**

Submitted to ARR

2025

- **Authors:** **Donggyu Lee**, Hyeok Yun, Meeyoung Cha, Sungwon Park, Sangyoon Park, Jihee Kim

RESEARCH EXPERIENCE

Stanford Doerr School of Sustainability, Stanford University, Stanford, CA, USA

Visiting Researcher

Jul 2026

- Adapted and enhanced satellite-based crop classification methods for rice, maize, and other major crops to improve transferability and performance in data-scarce, low-resource regions.
- Conducted research with Prof. David Lobell and Prof. Jennifer Burney on satellite-based crop mapping for North Korea, in collaboration with the Ministry of Unification of the Republic of Korea.

	Data Science for Humanity , Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany <i>Research Intern</i> Aug 2025 – Dec 2025 • Worked on satellite-imagery-based economic indicator estimation, integrating geospatial analytics with machine learning for socio-economic assessment. • PI: Prof. Meeyoung Cha (MPI-SP & KAIST)
	Applied Artificial Intelligence Lab , Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea <i>Undergraduate Researcher</i> Mar 2024 – Jun 2024 • Studied recent diffusion model literature and implemented custom diffusion model training pipelines. • PI: Prof. Ilchul Moon (KAIST)
	Financial Engineering Lab , Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea <i>Undergraduate Researcher</i> Jan 2021 – Feb 2021 • Contributed to a machine-learning-based financial product recommendation project, assisting with model design and implementation. • PI: Prof. Woochang Kim (KAIST)
WORK EXPERIENCE	Samsung Electronics , Suwon, South Korea <i>Machine Learning Intern</i> Jan 2024 – Feb 2024 • Developed diffusion-based defect image generation methods to enhance ML algorithms used in Galaxy smartphone quality assurance. • Selected as one of the top intern projects and delivered a final presentation to senior researchers and fellow interns.
	onCloud9 , Daegu, South Korea <i>Front-End Developer</i> Jan 2022 – Feb 2022 • Developed front-end components for the launch of a new blockchain-based service.
PROJECTS	Regional Economic Indicator Estimation with LLMs in European Countries <i>Collaboration with Samsung Electronics</i> Aug 2024 – Jul 2025 • Contributed to an end-to-end project estimating granular economic indicators (e.g., GRDP, labor force participation, purchasing power per capita) in four European countries using LLMs and satellite imagery, achieving Spearman correlations above 0.9 with official GRDP statistics. • Took primary responsibility for the satellite-to-LLM pipeline, including satellite tiling, geospatial preprocessing, and economic indicator prediction experiments with language models.
TEACHING EXPERIENCE	CS101: Introduction to Programming , KAIST, Daejeon, South Korea <i>Undergraduate Teaching Assistant</i> 2020 Fall; 2021 Spring/Fall; 2022; 2023; 2024 Spring • Assisted first-year students during lab sessions and provided hands-on programming guidance.
OTHER EXPERIENCE	NTU Summer School 2025 on AI for Finance , Nanyang Business School, Nanyang Technological University (NTU), Singapore <i>Participant</i> Jul 2025 – Jul 2025 • Participated in intensive lectures and workshops on AI and machine learning for finance, including financial forecasting, asset pricing, reinforcement learning, NLP, and large language models.
	Google Developer Student Club (GDSC) , KAIST, Daejeon, South Korea <i>Member</i> Mar 2023 – Aug 2023 • Joined the ML team, regularly reading and discussing research papers.
	KAIST Gifted Student Program , Daejeon, South Korea <i>Mentor</i> May 2021 – Nov 2021

- Mentored elementary-level students in an introductory programming camp, teaching basic coding concepts and problem-solving skills.

KAIST High School Mentoring Program, Daejeon, South Korea

Mentor

Jul 2021 – Jul 2021

- Provided academic guidance to high school students and shared insights on KAIST student life.

Republic of Korea Army, Korea

Sergeant

Mar 2019 – Oct 2020

- Served in a coastal surveillance unit near the Pohang region as part of mandatory military service.